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Deep Audio Detection Networks from Passive Acoustic Monitoring of a Critically Endangered Primate

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ABSTRACT

Biodiversity monitoring is a crucial component of conservation, providing essential information on species occurrence, population dynamics, community composition, and ecosystem structure. Recently, to enhance wildlife protection, passive acoustic monitoring (PAM) technology has been developed based on animal sound characteristics and has become an important tool for wildlife monitoring. The vast amount of data generated by PAM has been transformed by the "big data" revolution, and research methods that integrate artificial intelligence (AI) with efficient and powerful machine learning models are rapidly advancing. The primary challenge is to isolate the vocalization data of target species from this massive dataset and to determine whether the developed methods can be applied to other species. In this study, we focused on the acoustic signals of a critically endangered white-headed langur, a primate species endemic to China. Unlike traditional methods that transform sound frequencies into spectrogram images, our approach uses a Deep Audio Detection Network (DeepADN) that directly converts audio into acoustic features, which are then fed into a convolutional neural network for accurate detection of white-headed langur calls, even in noisy environments. Our method optimized detection performance, achieving a recall rate of 98.22% and reducing manual review workload by 87.07%. Furthermore, DeepADN demonstrated cross-species applicability by successfully detecting François' langur calls, highlighting its potential for broader conservation monitoring efforts.

1 | Introduction

Biodiversity monitoring is a key element of conservation, providing basic information on species occurrence, population dynamics, community composition, and ecosystem structure (Foden et al. 2013). However, ecology and conservation initiatives often involve wide spatial and temporal scales, which can be limited by practical circumstances and economic challenges (Bissonette 1999; Williams et al. 2018). Because longitudinal tracking of wildlife requires considerable manpower, material and financial resources, this artificial monitoring method is difficult for large land mammals, marine mammals, or some fast-moving species (Kalan et al. 2016). In addition, species monitoring using tracking devices or radio collars can be harmful to wildlife and is usually expensive (Murray and Fuller 2000). Therefore, to protect wildlife more effectively, especially some rare and endangered species,

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we urgently need to adopt efficient noninvasive monitoring methods in the field. Vocal communication is an important mode of interaction among animals, integral to maintaining social relationships (Fan et al. 2018; Ma et al. 2022), defending territory and food resources (Ma et al. 2020), attracting mates (Milich et al. 2014), and avoiding predators (Townsend et al. 2010). The saliency, flexibility, and potential for long-distance transmission, makes vocal communication one of the most effective communication methods (Rasmussen et al. 2024). Accordingly, passive acoustic monitoring (PAM) technology developed based on animal sound characteristics, has become an important means of wildlife monitoring (Rosenstock et al. 2002; Weir et al. 2005). The integration of artificial intelligence-driven frameworks and advanced computational models is now addressing the challenges posed by the massive and complex datasets generated through PAM, enabling unprecedented scalability and precision in wildlife vocalization studies (Dufourq et al. 2022; Stowell 2022; Tuia et al. 2022).

Significant advancements in animal sound recognition have been made in recent years, driven by the application of AI techniques. Convolutional neural networks (CNNs), in particular, demonstrate remarkable efficacy in various animal sound recognition tasks. Numerous studies have successfully employed CNNs to identify the calls of various animal species, including birds and cetaceans. Moreover, converting sound data into spectrograms at varying scales and using them as input for CNN classification has proven to be a highly effective approach (Bergler et al. 2019; Bermant et al. 2019; Stowell et al. 2019). Furthermore, data augmentation techniques, such as pitch and time shifting, along with the screening of background noise, have significantly enhanced model performance in recognizing target animal sounds (Bergler et al. 2019; Sprengel et al. 2016). Deep learning methods have also proven effective in processing large scale acoustic data, significantly reducing the workload of manual labeling and providing novel avenues for long-term ecological monitoring (Cauzinille et al. 2024). Kiskin et al. (2020) used the Continuous Wavelet Transform (CWT) (Flandrin 1998) instead of the Discrete Fourier Transform (Short-Time Fourier Transform, STFT) (Allen and Rabiner 1977) for generating spectrograms, a strategy that has proven highly effective. In the context of weak audio signals, Lakdari et al. (2024) demonstrated that in noisy environment, conventional Mel Frequency Cepstral Coefficients (MFCC) (Huang et al. 2001) outperform pre-trained neural networks in recognizing female northern gray gibbons (*Hylobates funereus* Geoffroy Saint-Hilaire, 1850) with respect to feature embedding and acoustic index computation. This finding (Lakdari et al. 2024) highlights the importance of considering specific application scenarios and environmental conditions when selecting acoustic feature extraction methods. To provide a clearer overview of current research and methods in this field, Table 1 summarizes studies using deep learning techniques for animal sound recognition. It includes information on the species, target sounds, methods, input features, and performance metrics like recall and F1 score.

Despite these advances, AI continues to face significant challenges in animal sound detection. Many studies rely on converting audio into images for detection, a process that inherently increases computational complexity and may result in the loss of certain details. In real-time scenarios, this conversion can introduce delays, making it incompatible with dynamic monitoring. Access to high-quality labeled data remains a major challenge, as many studies continue to depend on extensive manual labeling (Fairbrass et al. 2019). Additionally, transferring models between species remains problematic, as performance can be unstable across environments and background noise, with generalization capabilities needing improvement (Dufourq et al. 2022; Kiskin et al. 2020; Stowell et al. 2019). Training effective models for rare species or those with limited call data remains a substantial hurdle (Dufourq et al. 2021).

PAM has been widely used to monitor cetaceans, bats, birds, elephants, and anurans, but has been rarely applied to primates (Pérez-Granados and Schuchmann 2021; Sugai et al. 2019; Swider et al. 2022). As most primates are very vocal, PAM offers a very promising opportunity for primatological research. In addition, habitat fragmentation of many primates can make traditional monitoring methods challenging, which PAM can solve (Batist et al. 2024). In Vietnam, PAM has been successfully used to investigate the presence, distribution, and probability of occurrence of gibbons (Vu and Doherty 2021; Vu and Tran 2019). Using PAM to assess the activity of wild chimpanzees over a 24-h period, it was found that they have a diurnal pattern of calling with crepuscular peaks (predawn and after dusk) (Piel 2018). In addition, PAM has been successfully applied to explore the movement behavior, spatial distribution, and temporal and spatial characteristics of chimpanzees use and defense of territory (Kalan et al. 2016). Through the integration of PAM and machine learning, valuable new insights were gained into the behavior of the black-and-white ruffed lemur [*Varecia variegatac* (Kerr, 1792)] (Batist et al. 2024), including the previously unrecognized occurrence of nocturnal calls. In China, PAM has been applied in the study of two rare and endangered gibbons, Hainan gibbon [*Nomascus hainanusc* (Thomas, 1892)] and western black crested gibbon [*Nomascus concolorc* (Harlan, 1826)] (Dufourq et al. 2021; Zhong et al. 2021). Although the automatic detection approaches applied to the Hainan gibbons and black-and-white ruffed lemurs are commendable, there remains potential for further optimization. In both studies, audio data undergo feature extraction to obtain the Mel-spectrogram, which is subsequently transformed into an image for image-based detection. This approach inherently increases computational complexity, and the process of subsampling, the original audio to fit the image dimensions leads to a loss of critical audio information.

The white-headed langur (*Trachypithecus leucocephalus* Tan, 1955) is a critically endangered primate that lives in the karst mountains of Southwest Guangxi, China. They exhibit a strictly one-male and multi-female matrilineal social organization with an average group size of 12 individuals (Jin et al. 2009). At present, the population of white-headed langurs is only about 1400 individuals (Fan et al. 2024). It is difficult to monitor populations due to the heterogeneity and discontinuity of their habitat, as well as the complex topography of karst limestone forests (Tang et al. 2024). The latter situation is ideally suited to efficient PAM, combined with deep learning methods for population monitoring of this rare and endangered primate species endemic to China.

A previous study demonstrated that adult male white-headed langurs produce three distinct types of long-distance calls that are crucial for territorial defense and group coordination (Fan et al.

TABLE 1 | Summary of animal sound detection using deep learning.

Publications	Species	Methods	Input features	Recall (%)	F1 score (%)
Kalan et al. (2016)	Chimpanzee [*Pan troglodytes* (Blumenbach, 1775)]	Support vector machines or Gaussian mixture models	MFCC and loudness	11.0	5.8
Bermant et al. (2019)	Sperm whale (*Physeter macrocephalus* Linnaeus, 1758)	CNN	STFT-spectrogram	100	100
Bermant et al. (2019)	Killer whale [*Orcinus orca* (Linnaeus, 1758)]	CNN	STFT-spectrogram	93.77	93.02
Kiskin et al. (2020)	Southern house mosquito (*Culex quinquefasciatus* Say, 1823)	CNN	CWT-spectrogram	92.0	91.3
Shiu et al. (2020)	North Atlantic right whale [*Eubalaena glacialis* (Müller, 1776)]	Deep neural networks	STFT-spectrogram	94.6	73.43
Tzirakis et al. (2020)	Bornean gibbon (*Hylobates muelleri* Martin, 1841)	CRNN	Mel-spectrogram	93.3	—
Anders et al. (2021)	Chimpanzee [*Pan troglodytes* (Blumenbach, 1775)]	CRNN	Mel-spectrogram	—	33.0
Dufourq et al. (2021)	Hainan gibbon [*Nomascus hainanus* (Thomas, 1892)]	CNN	Mel-spectrogram	96.48	90.55
Lapp et al. (2021)	Boreal chorus frog [*Pseudacris maculata* (Agassiz, 1850)] and Variable harlequin frog (*Atelopusvarius* Lichtenstein and Martens, 1856)	Repeat interval-based bioacoustic identification tool	Pulse repetition interval	56.0	69.0
PérezGranados and Schuchmann (2021)	Black and gold howler monkey [*Alouatta caraya* (Humboldt, 1812)]	Kaleidoscope Pro (an automated signal recognition software)	STFT-spectrogram	89.2	—
Bergler et al. (2022)	11 species	CNN	STFT-spectrogram	97.4	97.3
Wang et al. (2022)	Hainan gibbon (*N. hainanusc*)	CRNN	Mel-spectrogram	≈90.0	≈90.0
Bonafos et al. (2023)	Guinea baboon [*Papio papio* (Desmarest, 1820)]	CNN	Mel-spectrogram	90.28	82.68
Lawson et al. (2023)	Geoffroy's spider monkey (*Ateles geoffroyi* Kuhl, 1820)	CNN	Mel-spectrogram	75.0	62.0
Ravaglia et al. (2023)	Indri [*Indri indri* (Gmelin, 1788)]	CNN	STFT-spectrogram	86.0	68.0
Frainer et al. (2023)	Indian Ocean humpback dolphin [*Sousa plumbea* (Cuvier, 1829)]	CNN	STFT-spectrogram	56.7	68.9
Batist et al. (2024)	Black-and-white ruffed lemur [*Varecia variegata* (Kerr, 1792)]	CNN	Mel-spectrogram	94.0	70.0
Bota et al. (2024)	American bullfrog [*Lithobates catesbeianus* (Shaw, 1802)]	BirdNet	Mel-spectrogram	89.5	93.9
Holderried et al. (2024)	Eurasian Woodcock (*Scolopax rusticola* Linnaeus, 1758)	BirdNet	Mel-spectrogram	89.7	—
Kath et al. (2024)	42 species of anurans	BirdNet	Mel-spectrogram	—	59.1

Note: – indicates that the data is not available in the reference.

2022). The aim of this study is to develop an acoustic feature-based automatic audio detection model using these three well-documented long-distance calls, to enhance conservation efforts for this rare and endangered species. The main contributions of this work are as follows: First, we develop a Deep Audio Detection Network (DeepADN) to evaluate the effectiveness of using acoustic features and CNNs to detect white-headed langur calls from PAM data. Second, we compare the performance of various deep learning architectures combined with acoustic features, for automatically identifying these specific calls and explore strategies to optimize the performance of automatic detection models. We use recall as the optimization metric, as automatic detection is the first step in PAM. Only by identifying as many white-headed langurs' calls as possible can subsequent domain analysis, individual identification, behavioral studies, and other research be conducted. Finally, we demonstrate the transfer ability of our model to other endangered primate monitoring tasks. The methodological framework established in this study is expected to be a valuable template for sound monitoring of other endangered primate species, particularly those inhabiting complex terrains, such as the Hainan gibbons (*N. hainanus*) and eastern black crested gibbons [*Nomascus nasutus* (Kunkel d'Herculais, 1884)]. By establishing a robust and adaptable method for automated call detection, this research contributes to the broader field of primate conservation technology, addressing the urgent need to conserve white-headed langurs.

2 | Materials and Methods

2.1 | Study Site and Subjects

The study site is located in the Banli area of the Chongzuo White-Headed Langur National Nature Reserve in Guangxi (22°10′ to 22°17′N, 107°29′ to 107°31′E). The region experiences a tropical monsoon climate, characterized by hot, rainy summers and dry, less rainy winters. The average annual rainfall is approximately 1200 mm, with an average annual temperature of around 21.6°C (Yang et al. 2022). The reserve is dominated by rugged karst limestone hills, with elevations ranging from 400 to 600m (Zhou et al. 2010). The valleys within the study area have been extensively planted with sugar cane, resulting in drastic fragmentation of the white-headed langur's forest habitat (He et al. 2021).

The study focused on 11 habituated white-headed langur groups and several unstudied groups within the reserve. Because white-headed langurs exhibit a strictly one-male and multi-female matrilineal social organization, each group consisted of one adult male and multiple adult females, as well as several juvenile offspring, during the study period. Our primary focus was on the long-distance calls of these adult male langurs.

2.2 | Audio Data Collection

2.2.1 | Manual Data Collection

From April 2020 to August 2023, 11 habituated groups of langurs were observed at the rate of one group per day for 15–25 days per month to ensure adequate data in both the dry season (October to March of the next year) and the wet season (April to September). Observations were conducted from 5:30 a.m. to 8:00 p.m. in the wet season and from 6:00 a.m. to 7:00 p.m. in the dry season. We used the continuous focal animal sampling method to track male langurs. We used a Zoom F6 digital sound recorder (Zoom, Japan) with a Rode NTG8 directional microphone (Rode, Australia) or a Sennheiser ME66 directional microphone (Sennheiser, Germany) to collect audio data at a 44.1 kHz (16 bits) sampling rate. Whenever an adult male langur emitted a long-distance call, we manually started the recording device to ensure that the entire call was recorded before turning it off. Meanwhile, we carefully recorded the start time of each call, the identity of the vocalizing individual, and the context in which the call occurred.

2.2.2 | PAM Data Collection

We use 12 solar-powered Autonomous Recording Units (ARUs; Lindon Ecology, Model LBird- 01211) to passively record calls. From September to December 2023, we placed the devices in the range of three habituated groups (four devices each) of white-headed langurs, with deployment locations illustrated in Figure 1. Between January and March 2024, we deployed devices in areas where unknown groups were likely to be present; these locations are also shown in Figure 1. Each device was installed on a metal pole 3–4m above the ground, with a minimum distance of 200m between units. The distance ensures no overlap in acoustic coverage, as each unit's effective recording radius is 100m in dense forests. The ARUs had a sampling rate of 32 kHz, recording in 16 bit mono and MP3 format. Recordings were made during the langurs' active vocalizing periods, which were from 5:00 a.m. to 12:00 p.m. and 3:00 p.m. to 8:00 p.m. each day (Fan et al. 2022). Throughout these time frames, the automatic recording units functioned continuously without any interruption. Therefore, when the devices were operating normally, 12h of recording data were generated daily. To ensure complete capture of call sequences, the ARUs were programmed to generate nonoverlapping 12 s recordings. These recordings were triggered continuously and consecutively, with each new file commencing immediately after the prior segment concluded, thereby eliminating gaps or overlaps between files. Detailed descriptions of call sequences are provided in the Supporting Information S1. We checked the ARUs every 2–3 weeks.

The calls of white-headed langur are of low frequency and not particularly loud. Unlike high-frequency species like bats or louder primates like gibbons, white-headed langur's calls are disproportionately impacted by anthropogenic noise due to their low-frequency structure. In their sugarcane plantation-adjacent habitat, critical interference arises from human speech, agricultural machinery, and manual labor activities, all of which spectrally overlap with langur calls, accelerating signal degradation.

To quantify the acoustic parameters of langur calls and ambient noise levels, field measurements were conducted using the SMART SENSOR AR814 sound level meter (SMART SENSOR, China) and the SNDWAY SW-600A rangefinder (SNDWAY, China). The long-distance calls of white-headed langurs were recorded at a distance of approximately 80 m from the vocalizing individual, with sound levels ranging from 48.2 to 54.4 dB. At the

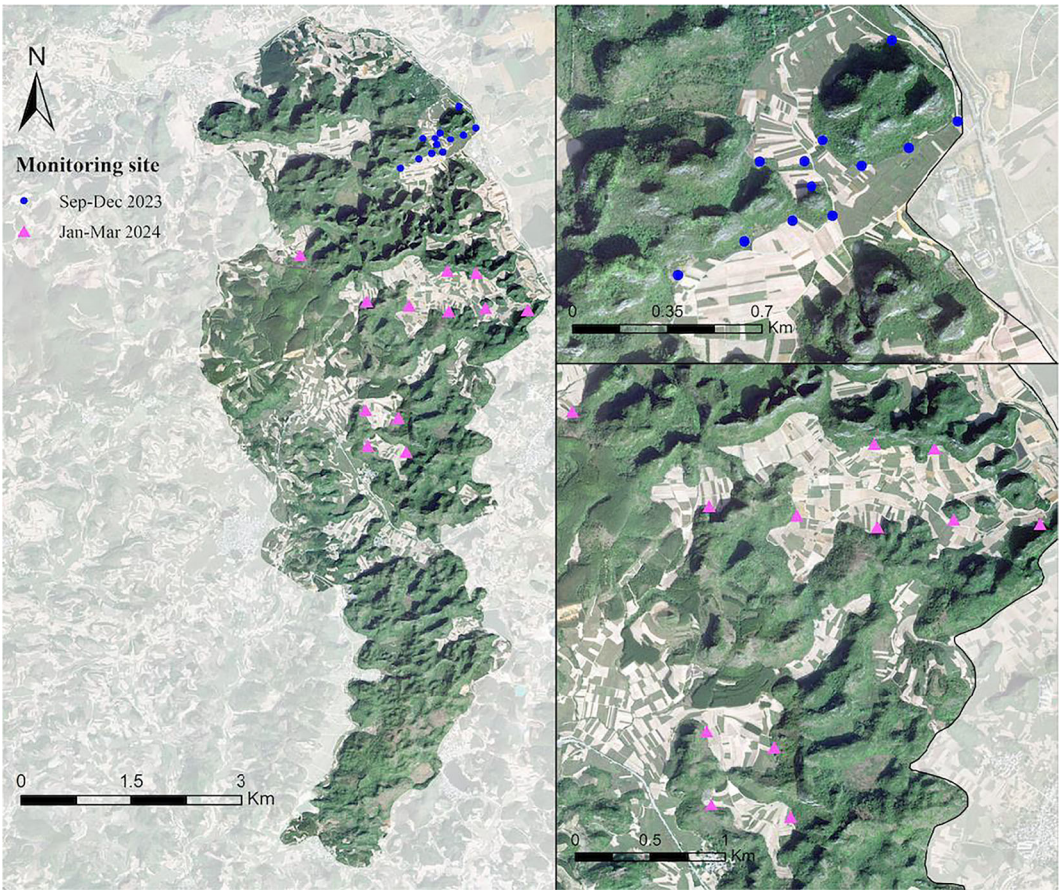

FIGURE 1 | The monitoring sites deployed within the study area (Banli area, Chongzuo White-Headed Langur National Nature Reserve, Guangxi, China). Satellite imagery was obtained from ESRI World Imagery.

conclusion of each call, the environmental noise at the location of the vocalizing individual was measured, yielding values between 38.2 and 41.9 dB.

Initial noise reduction attempts using spectral gating (via Python package noise reduce) revealed risks of signal distortion due to spectral overlap between langur calls and anthropogenic noise. The signal-to-noise ratio (SNR) was calculated from the raw data using the same spectral gating framework, yielding an average SNR of −1.68 dB. However, this value likely overestimates the true SNR, as residual noise from overlapping frequency bands could not be fully isolated. Consequently, all subsequent call detection analyses were conducted on unprocessed recordings to preserve vocal fidelity, avoiding potential biases introduced by noise reduction algorithms. While this approach retains ambient noise, it ensures the integrity of the original vocal signal.

2.3 | Audio Data Management

We collected a total of 498.6 h (143 016 files) of audio data, which included 24.6 h (816 files) of manually recorded audio, with the remainder 474.0 h (142 200 files) sourced from the ARUs. To ensure consistency across audio formats, we resampled the manually recorded audio from its original 44.1 kHz sampling rate to match the 32 kHz rate of the ARUs using the Sinc algorithm (Stenger 2000). Additionally, we converted the audio files from WAV to MP3 format. ARUs recordings were standardized into 12 s segments, and we applied the same segmentation method to the manually recorded calls, using a sliding window of 12 s with no overlap. For any final segment shorter than 12 s, we padded it with zeros to extend the length to the required duration.

Consistent with previous studies (Fan et al. 2022), our research confirmed that the calls of adult male white-headed langurs predominantly occur within the 0–3 kHz frequency range. Therefore, we first applied a low pass filter to all audio data, retaining only frequencies within this range. We then manually classified the filtered audio files by auditory and visual inspection into calls of adult male white-headed langurs (positive; Figure 2a), separate from recordings lacking these calls (negative). To characterize the acoustic properties of the negative class while maintaining computational efficiency, we performed a preliminary manual categorization of the negative class through auditory and spectrogram-based visual inspection, focusing on features representative of the local acoustic environment. The negative class was further subdivided into three classes: natural environmental sounds (Figure 2b), bird vocalizations (Figure 2c), and other biological sounds (Figure 2d). All classifications were performed independently by four authors, followed by cross-validation to ensure consistency and consensus, resulting in the final annotated dataset. The negative class exhibited greater internal variability and less distinct boundaries. To enhance the model's ability to learn the acoustic features of each subclass,

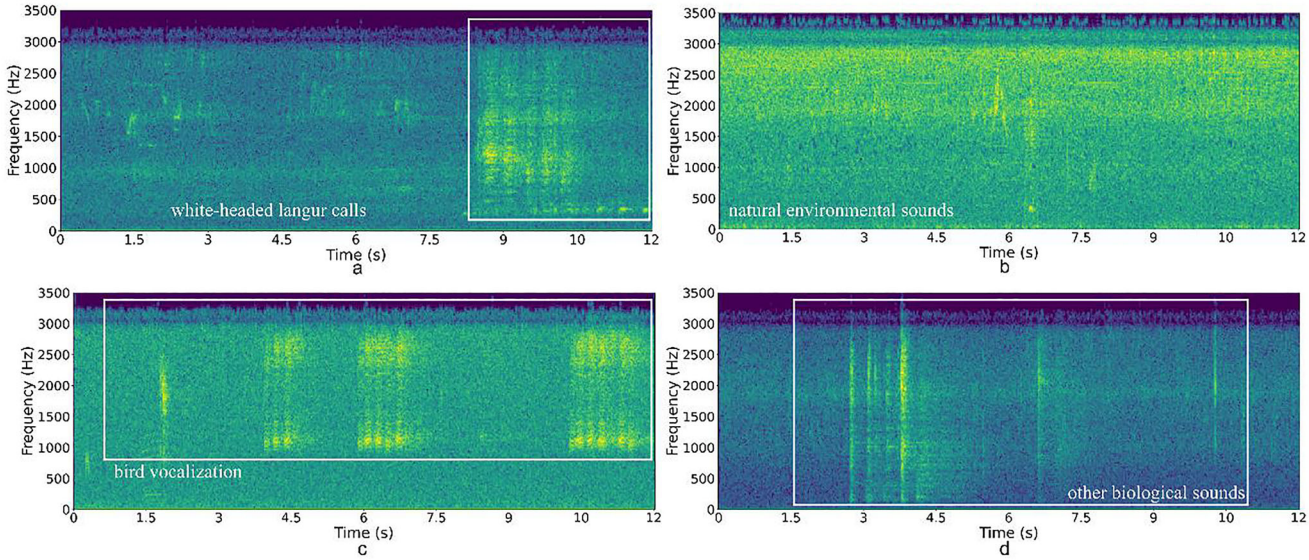

FIGURE 2 | Examples of manual annotation. The figure contains four audio spectrograms: (a) adult male white-headed langur calls, (b) natural environmental sounds, (c) bird vocalizations, and (d) other biological sounds.

TABLE 2 | Dataset.

Dataset	Number of training set		Number of testing set	
	Negative	Positive	Negative	Positive
PAM	13,434	1575	5758	675
Manual	4213	853	5758	675
Mix	13,434	2428	5758	675
PAM-4	13,434	6300	5758	675
Manual-4	16,852	3412	5758	675
Mix-4	13,434	9712	5758	675

Note: All testing sets are derived from the ARUs recordings. PAM refers to the training set recorded by ARUs; Manual refers to the training set composed of manually recorded data; Mix is the training set that combines positive class audio from the Manual set with the PAM training set. PAM-4 quadruples the augmentation of the positive class in the PAM training set, Manual-4 quadruples the augmentation of the Manual training set, and Mix-4 quadruples the augmentation of the positive class in the Mix training set.

we increased the number of samples in the negative class. Since ARUs capture larger samples, each negative subclass should be 2–3 times the total of the positive class to ensure adequate coverage of the dataset.

We applied data augmentation techniques, including time shifting (Cui et al. 2015), time-domain masking (Ko et al. 2017), and frequency-domain masking (Park et al. 2019), to implement resampling (Batista et al. 2004) of the positive class in the training set. Additionally, we combined manually recorded positive class vocalizations with ARUs recorded sounds to further enhance the diversity of the data (Table 2).

2.4 | Evaluation Metrics

To identify adult male white-headed langur calls, accurate detection and identification of all positive samples is crucial. Therefore, our primary evaluation metric is "recall," which, unlike accuracy (the ratio of the number of correct predictions to the total number of predictions), focuses on the proportion of positive samples correctly identified, providing a better measure of the model's ability to detect positive events. In addition, we assign the F1 score as a secondary evaluation metric, combining precision and recall through their harmonic mean to reflect the model's performance in reducing the event cost of manual review by minimizing false positives and false negatives, ensuring accurate and cost-efficient identification of positive samples. The definitions of these evaluation metrics are as follows:

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{1}$$

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{2}$$

$$\text{F1 score} = \frac{2 \times \text{recall} \times \text{precision}}{\text{recall} + \text{precision}} \tag{3}$$

True positives (TP) represent the number of genuinely positive samples that are correctly predicted as positive. False positives

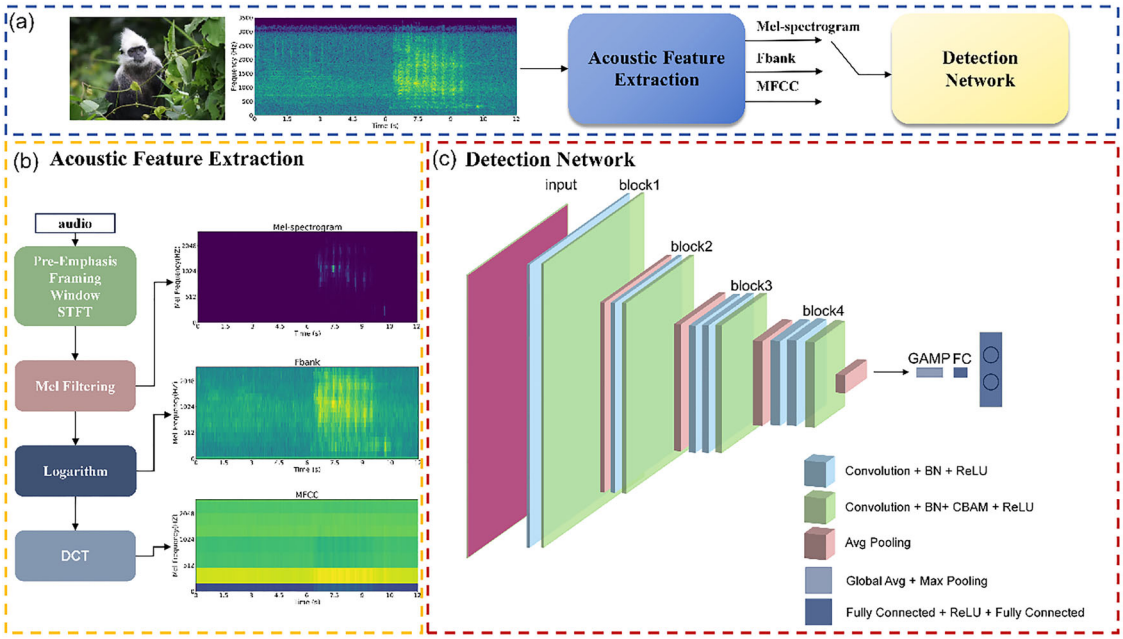

FIGURE 3 | Illustration of steps in applying the DeepADN model. The process begins with (a) the overall architecture, followed by (b) the acoustic feature extraction module, and concludes with (c) the audio detection network.

(FP) refer to the number of samples that are actually negative but incorrectly predicted as positive, often referred to as false alarms. False negatives (FN) denote the number of genuinely positive samples that are incorrectly predicted as negative, also known as misses.

2.5 | DeepADN

We developed a custom audio detection model, DeepADN, tailored to detect sounds based on their acoustic features. The architecture of DeepADN is shown in Figure 3a. The model comprises two main components: an acoustic feature extraction layer and an audio detection network. Pre-processed audio is fed directly into the model, where it passes through both components, resulting in a detection evaluation of whether or not the audio contains calls of adult male white-headed langurs.

2.5.1 | Acoustic Feature Extraction

Our acoustic feature extraction process generated three different audio feature vectors, namely, Mel-spectrogram (Rabiner & Juang 1993), filter bank energy (Fbank) (Biem et al. 2001), and MFCC (Huang et al. 2001). One of these acoustic features is then passed as input to the feature processing layer (Figure 3b; Supporting Information S2).

The Mel-spectrogram captures the energy distribution across frequencies using the Mel scale, which aligns with human auditory perception by emphasizing lower frequencies. It is commonly derived through steps like pre-emphasis, STFT (Allen and Rabiner 1977), and applying a Mel filter bank. Mel-spectrogram is widely used in speech recognition, audio classification, and music information retrieval to capture perceptually relevant features of sound.

Fbank reflects the signal's spectral content by measuring energy across frequency bands through a series of Mel filters, which are applied to the audio spectrum to capture important frequency characteristics; this process not only provides insights into the timbre and texture of the signal but also generates features, referred to as filter bank coefficients, that are widely used in speech recognition and other audio analysis tasks to enhance the understanding and classification of complex sounds.

MFCC combines the Mel-spectrogram with a discrete cosine transform (DCT) (Ahmed et al. 1974) to focus on the spectral envelope and formants, capturing the perceptually important characteristics of speech. Specifically, MFCC is derived by applying a Mel filter bank to the frequency spectrum, followed by a DCT to reduce dimensionality. MFCC captures the perceptually important characteristics of speech and is widely used in speech recognition, speaker identification, and audio classification tasks.

2.5.2 | Audio Detection Network

We evaluated the audio detection capabilities of various networks using the PAM dataset, including Ecapa-TDNN (Desplanques et al. 2020), PANN (Kong et al. 2020), Res2Net (Gao et al. 2019), ResNet (Targ et al. 2016), CAMPPlus (Wang et al. 2023), and ERes2NetV2 (Chen et al. 2023). Among them, PANN stands out in terms of its superior performance. We further compared the PANNs-CNN6, PANNs-CNN10, and PANNs-CNN14. The detection network architecture in DeepADN is shown in Figure 3c.

The four-convolutional-block structure of PANNs-CNN10 was retained, with additional convolutional layers added to the third and fourth blocks to enhance the model's performance and accuracy in recognizing adult male white-headed langur calls. In the original model, each block contained two convolutional layers. In our modified version, the first two blocks remain unchanged, each with two convolutional layers, while the third and fourth blocks were expanded to include three convolutional layers each. All convolutional layers follow a consistent structure: a 3×3 convolutional layer, a batch normalization layer, and an ReLU activation function. More details of convolutional layers can be found in the Supporting Information.

To enhance model performance, we incorporated the lightweight Convolutional Block Attention Module (CBAM), which improves feature representation (Woo et al. 2018). CBAM comprises two submodules: the Channel Attention Module (CAM) and the Spatial Attention Module (SAM), which prioritize relevant information along the channel and spatial axes, respectively (see Supporting Information for more details of the CBAM). In deep networks, the input feature distribution can shift significantly at each layer due to parameter updates, resulting in variation in statistical properties across data batches. If the feature distributions entering the attention mechanism are unstable or overly diverse, attention weights may be inaccurately calculated. To mitigate this issue, we inserted CBAM between the batch normalization layer and the ReLU activation function. The batch normalization layer stabilizes the feature distribution before it enters CBAM, reducing discrepancies in numerical ranges and allowing CBAM to better focus on the relative importance of features. This not only improves model performance but also enhances stability. By placing CBAM before the activation function, the network can directly weight the output features from the convolutional layer, ensuring that more important features receive higher responses during activation while preventing suppression caused by the nonlinearity of the activation function.

In audio data, irrelevant or redundant information can lead the network to focus on noise, hindering the extraction of key features and resulting in overfitting. To address this, we employed Dropout regularization, which randomly sets the output of certain nodes to zero during each training step with a specified probability (Srivastava et al. 2014). These selected neurons do not participate in computations or parameter updates, encouraging the model to use different "sub-networks" in each training session. This randomness reduces complex co-adaptations between neurons, helping the model learn more robust feature representations, improving generalization, and reducing overfitting. Finally, the global average pooling layer extracts important features, which are then passed through the first fully connected layer followed by an ReLU activation function (He et al. 2016). This combination introduces nonlinearity, enhancing the model's expressive power. The second fully connected layer maps the features to the two output categories.

2.6 | Validation of Transfer Learning

To evaluate DeepADN's adaptability to other species, we applied transfer learning techniques to adapt a pre-trained call detection model for adult male white-headed langurs to a limited-sample dataset of adult male François' langurs (*Trachypithecus francoisi* Pousargues, 1898) (Xie et al. 2024), recorded in captivity (Supporting Information S3).

As a closely related species of white-headed langurs, François' langurs also have age–sex-specific call types (Xie et al. 2024). François' langurs can produce a total of 23 call types, with adult males exhibiting five unique call types, and similar to white-headed langurs can produce three long-distance calls (Figure 4). When there is conflict or competition between groups of langurs, adult male langurs emit long-distance calls and remain alert. These long-distance calls have been hypothesized to have the ability to regulate inter-group relationships, maintain group cohesion, attract mates, defend the territory, and play an important role in male competition and maintenance of population survival and reproduction (Poirier-Poulin and Teichroeb 2020; Kitchen 2004).

We specifically designed a dataset for François' langurs, consisting of 130 manually annotated call samples with varying audio lengths. To create a dataset that reflected real-world conditions, we selected 4213 audio samples from the manual dataset that do not contain white-headed langur calls, which were used as negative samples. This dataset was then split into training and validation sets in a 7:3 ratio, ensuring sufficient data for both model adaptation and validation. Additionally, we applied quadruple augmentation to the positive class samples in the training set.

For model adaptation, we employed transfer learning by freezing the early layers of the pre-trained model and fine tuning only the later layers of the François' langur dataset. This approach preserved the foundational features learned from the original species while enabling the model to adapt to the species-specific characteristics in the upper layers. To prevent overfitting on the limited data, we applied a reduced learning rate during fine tuning (Supporting Information S4.1).

3 | Results

3.1 | Comparison of Basic Models

To validate the effectiveness of various acoustic features and network models for white-headed langur call recognition, and to compare their performance with image classification models, we evaluated different combinations on the PAM dataset (Table 3). Overall, models leveraging Mel-spectrogram, Fbank, and MFCC features outperformed those relying on Mel-spectrogram image inputs, highlighting the advantages of these acoustic features in primate call detection. PANNs-CNN10 achieved the best results with Fbank, showing a recall of 95.26% and an F1 score of 94.41%, illustrating Fbank's robustness in handling noisy environments. VGG16 (Simonyan 2014) similarly performed well with Mel-spectrogram and Fbank, achieving recall and F1 scores of 93.48% and 93.83%, respectively, both of which exceeded results of the Mel-spectrogram image. Although EfficientNet-B0 (Tan and Le 2019) and Vision Transformer (Dosovitskiy 2020) also showed strong scores with Mel-spectrogram image, models using Mel-spectrogram, Fbank, or MFCC all demonstrated higher scores. In contrast, DenseNet161 (Huang et al. 2017) displayed consistently

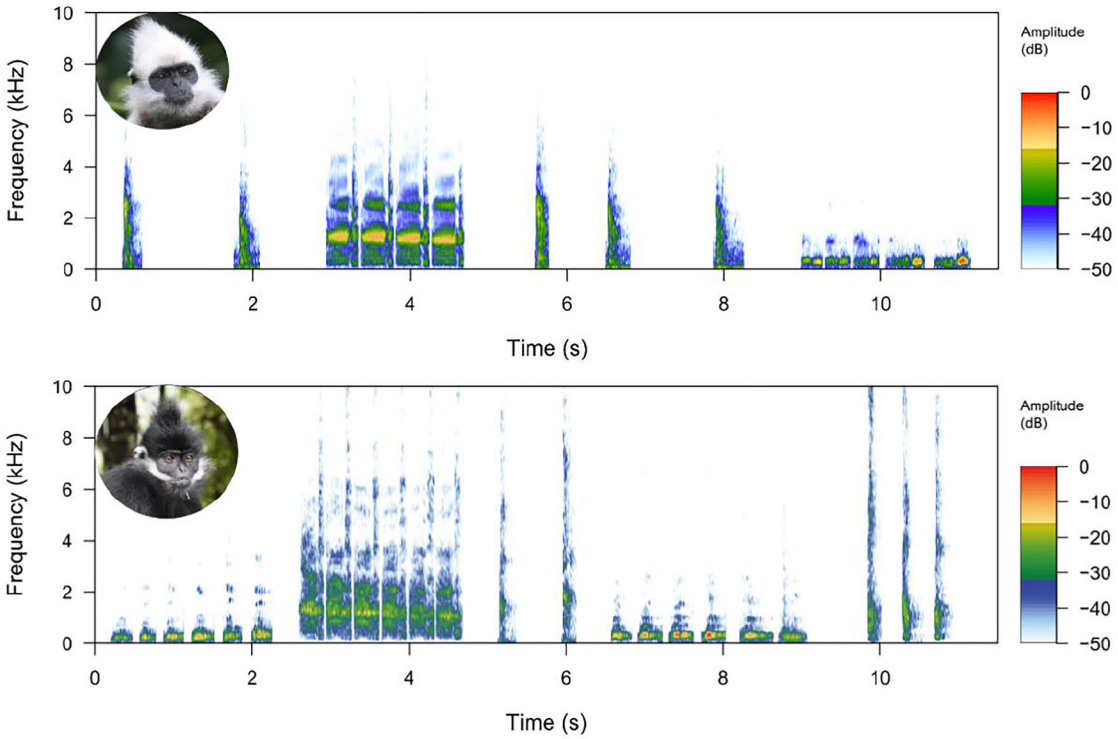

FIGURE 4 | Long-distance calls of white-headed langurs and François' langurs.

lower performance across all feature types, with notably reduced recall and F1 score. These findings underscore the importance of careful input feature selection and network architecture choice to optimize detection accuracy in PAM. The specific environmental configurations for all experiments are shown in Supporting Information S4.2.

3.2 | Performance of DeepADN

We conducted ablation experiments on the DeepADN model using Fbank features on the PAM dataset to evaluate the effect of adjusting convolutional block depth and incorporating CBAM. The experimental results demonstrate that the DeepADN model without CBAM achieved a recall of 96.15% and an F1 score of 93.25%, which was significantly improved compared to the original PANNs-CNN10 model (with a recall of 95.26% and an F1 score of 94.41%). Furthermore, the complete DeepADN model outperformed its version without CBAM, achieving a recall of 97.48% and an F1 score of 88.56%. This indicates that incorporating CBAM into DeepADN effectively enhances the model's performance. Overall, DeepADN reduces the number of missed positive samples from 32 (PANNs-CNN10) to 17.

We then compared the effect of different frame lengths and overlap lengths of the Fbank feature on DeepADN's performance using the PAM dataset (Table 4). Among the tested configurations, a frame length of 25 ms and an overlap of 10 ms achieved the best balance, with a recall of 97.48% and an F1 score of 88.56%.Other configurations, such as 10 ms frame length with 5 ms overlap and 30 ms frame length with 15 ms overlap, demonstrated varying levels of performance, but none surpassed the 25 and 10 ms combination. The Mix dataset achieved the highest scores with a recall of 98.22% and an F1 score of 88.81%, closely followed by the Mix-4 dataset, which had the same recall but a slightly lower F1 score of 86.72% (Table 5). The PAM and PAM-4 datasets do not perform as well as Mix and Mix-4, but they are relatively normal. In contrast, the Manual dataset has a recall of 93.78% but the lowest F1 score of 37.28%, while its enhanced version, Manual-4, has improved but still has a poor recall and F1 score.

3.3 | Five-Fold Cross-Validation

We conducted a five-fold cross-validation (Kohavi 1995) on the PAM, Mix, PAM-4, and Mix-4 datasets using DeepADN. The PAM dataset achieved an average recall of 97.21% and F1 score of 89.50%, while the Mix dataset showed an average recall of 97.96% and F1 score of 86.73%. The PAM-4 dataset yielded an average recall of 98.43% and an F1 score of 85.91%. The Mix-4 dataset achieved the highest stability, with an average recall of 98.55% and an F1 score of 87.44% (Figure 5). Due to poor performance, the Manual and Manual-4 datasets were excluded from cross-validation.

3.4 | Transfer Learning of the DeepADN

We performed transfer learning on the François' langurs dataset, with Fbank features yielding the best performance (Supporting Information S4.3). For the Fbank's performance, the result revealed the model's strong performance in detecting positive cases, with a true negative count of 1245, 39 true positives, 19 false positives, and no false negatives. Despite the challenging dataset,

TABLE 3 | Comparison of individual networks with different input features.

Model	Input feature	Recall (%)	F1 score (%)
EfficientNet-B0	Mel-spectrogram image	90.82	90.41
Vision Transformer	Mel-spectrogram image	61.48	70.70
DenseNet161	Mel-spectrogram image	9.78	17.48
VGG16	Mel-spectrogram image	55.56	67.39
	Mel-spectrogram	92.89	92.89
	Fbank	93.48	93.83
	MFCC	90.82	92.89
CAMPPlus	Mel-spectrogram	86.67	88.84
	Fbank	93.33	94.17
	MFCC	90.22	89.04
Ecapa-TDNN	Mel-spectrogram	73.82	82.57
	Fbank	92.89	93.86
	MFCC	73.82	82.57
ERes2Net	Mel-spectrogram	94.07	93.52
	Fbank	93.78	94.90
	MFCC	92	93.81
Res2Net	Mel-spectrogram	82.52	88.20
	Fbank	92.18	93.82
	MFCC	90.22	92.48
ResNet	Mel-spectrogram	91.85	93.23
	Fbank	92.59	93.83
	MFCC	93.33	93.33
PANNs-CNN6	Mel-spectrogram	92.44	90.11
	Fbank	95.26	93.87
	MFCC	92.74	91.32
PANNs-CNN10	Mel-spectrogram	92.89	92.00
	Fbank	**95.26**	**94.41**
	MFCC	89.33	89.33
PANNs-CNN14	Mel-spectrogram	93.48	92.18
	Fbank	94.96	94.68
	MFCC	92.74	92.33

TABLE 4 | Performance comparison of DeepADN with different frame and overlap parameters for Fbank features.

Parameters	Recall (%)	F1 score (%)
Frame 10ms, Overlap 5 ms	97.13	88.87
Frame 20 ms, Overlap 10ms	97.19	87.82
Frame 25 ms, Overlap 10 ms	**97.48**	88.56
Frame 25 ms, Overlap 12.5 ms	96.44	91.63
Frame 30ms, Overlap 15ms	96.89	90.52

Note: The bold item is the highest value in the table.

TABLE 5 | Performance comparison of DeepADN using different datasets.

Dataset	Recall (%)	F1 score (%)
PAM	97.48	88.56
Manual	93.78	37.28
Mix	**98.22**	**88.81**
PAM-4	97.78	85.83
Manual-4	74.67	58.84
Mix-4	**98.22**	86.73

Note: Bold items are the highest values in the table.

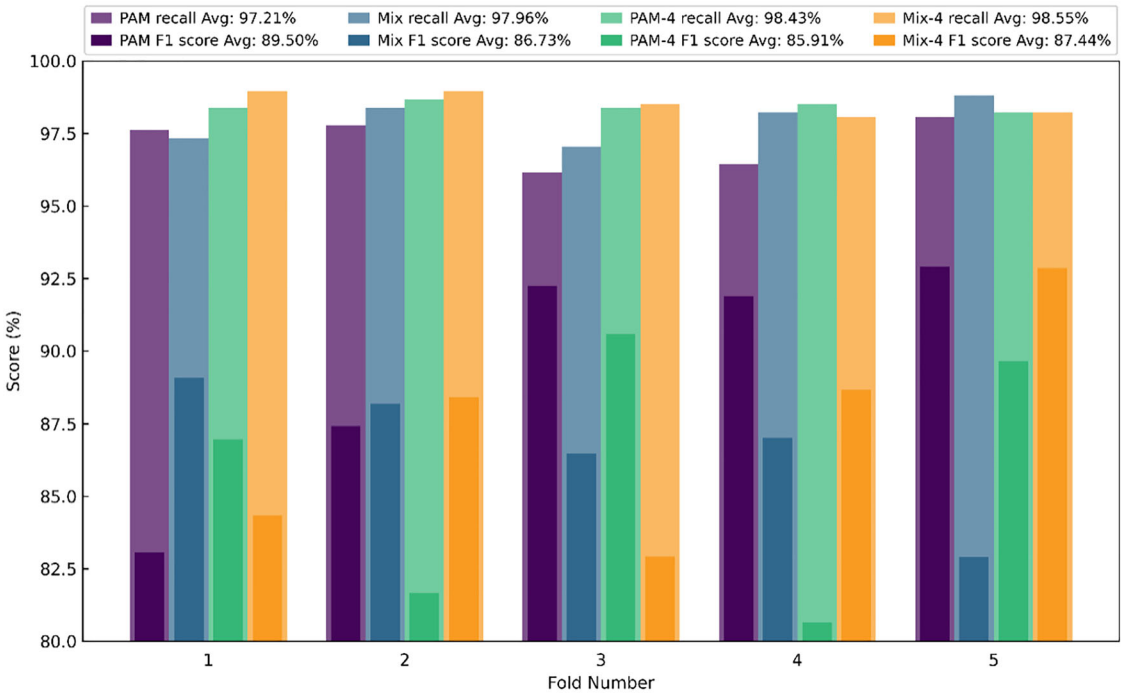

FIGURE 5 | Performance of recall and F1 results in five-fold cross-validation.

which is heavily imbalanced with a large number of negative samples, the model achieved an accuracy of 98.54% and a perfect recall of 100%, successfully identifying all positive samples.

4 | Discussion

4.1 | Fbank Is Better

We compared our audio detection network method to several recent studies focusing on acoustic monitoring of endangered primates. We demonstrate the potential of deep learning techniques for detecting and classifying primate calls. These studies, targeting species such as the Hainan gibbon (Dufourq et al. 2021), the black-and-white ruffed lemur (Batist et al. 2024), and the Indri (Ravaglia et al. 2023), provide useful insights into the application of machine learning to conservation.

A notable similarity across these works is the use of images to input features for neural networks analysis, whether these features are STFT-spectrogram or Mel-spectrogram. However, spectrogram images have limitations. Specifically, when sound is converted into a fixed-size image, longer segments often require subsampling compared to shorter ones in order to fit the image dimensions, which inevitably results in information loss and compromises the fidelity of feature extraction. As a result, spectrograms are most effective when applied to short, standardized-length audio clips. The typical duration of adult male white-headed langur calls is between 0.2 and 10 s (Fan et al. 2022), which, unfortunately, is not ideal for image conversion. This is because longer durations may lead to the loss of crucial acoustic details when forced into fixed-dimension spectrogram images. This limitation explains the relatively lower performance observed in our image-based models, such as VGG16 and DenseNet161, when compared to models leveraging acoustic features like Mel-spectrogram, Fbank, or MFCC. Compared to other studies, our set audio segment length is the longest, at 12s, which perfectly accommodates the calls of adult male white-headed langurs, preventing the truncation of continuous calls and laying the foundation for subsequent behavioral studies.

On the other hand, Mel-spectrogram features effectively capture the raw energy distribution of each timeframe across the Mel frequency bands, which reflect the frequency information inherent in the audio. However, they lack inherent adaptability to volume fluctuations. As a result, when the audio volume varies significantly, the Mel-spectrogram representation can become unstable, with feature values potentially fluctuating unpredictably. This instability can hinder the model's ability to consistently extract reliable features from recordings with varying volume levels. In contrast, Fbank features are computed using a logarithmic transformation, which has been proven to be a highly effective operation (Choi et al. 2018). This transformation compresses the dynamic range of sounds and enhances the stability of the extracted features. The logarithmic transformation allows Fbank features to maintain a consistent representation across different volume levels, making them more adaptable to audio data with diverse volume fluctuations.

MFCC features employ a DCT to remove redundancy and reduce computational complexity, which makes them particularly effective for short-duration audio analysis. However, for longer audio sequences, the high compression induced by DCT can lead to the loss of high dimensional frequency information, particularly those vocalizations with complex frequency components. In contrast, Fbank features, by avoiding excessive dimensional compression, retain more frequency information, making them better suited for detecting longer audio sequences and for handling complex acoustic soundscapes, such as the situation of multiple sound sources simultaneously emitting sound in natural environments.

4.2 | DeepADN Optimization Strategies

We first introduced the CBAM module (Woo et al. 2018), which significantly improved the base model's performance. By incorporating channel and spatial attention mechanisms, CBAM enables the network to focus on the most relevant features, enhancing detection accuracy. To further improve the precision and robustness of call detection, and make DeepADN more effective in capturing key elements in audio signals, especially in noisy environments, we optimized the parameter selection of the Fbank feature by choosing a frame length of 25 ms and an overlap of 10 ms, which achieved the best performance on the PAM dataset, with a recall score of 97.48% and an F1 score of 88.56%. Comparing various data augmentation techniques across different datasets, the Mix dataset achieved the highest recall score, followed closely by the Mix-4 dataset, which had the same recall but a slightly lower F1 score. Five-fold cross-validation (Kohavi 1995) further demonstrates the stability and validity of the model, with the Mix-4 dataset achieving the highest average recall as well as the second-highest F1 score of 98.55% and 87.44%, respectively. The comparison between PAM-4 and PAM highlights the crucial role of resampling in improving model performance for call detection, particularly by addressing class imbalance within the dataset. By expanding the number of positive samples through quadrupled augmentation, the PAM-4 dataset achieved more reliable detection accuracy, demonstrating that resampling (Batista et al. 2004) can effectively mitigate the limitations posed by underrepresented classes. Furthermore, the Mix-4 dataset, which combined targeted data augmentation with resampling, achieved even better recall and F1 scores than both PAM-4 and Mix alone. We recommend combining resampling with other data enhancement techniques, as they not only help balance the dataset but also improve the model's capacity to accurately recognize and distinguish endangered primate calls, even if data are imbalanced.

4.3 | Ability of DeepADN

For the testing set, the DeepADN model achieves a recall of 98.22% and a precision of 81.13%. When recall approaches 1, the workload is approximately the original workload multiplied by $\frac{1}{p(1+k)}$, where p is the precision and k is the ratio of negative samples to positive samples. This means that with the use of the model, the amount of manual review is only 12.93% of the original workload. The Hainan gibbon study has the capacity to achieve or even exceed such recall and accuracy rates, however, it would necessitate the implementation of additional manual post-processing by the observer following automated detection to achieve this (Dufourq et al. 2021).

One of the key challenges in applying DeepADN across different species is the imbalance in dataset composition, particularly when most of the samples belong to the negative class, and even a small number of false positives can significantly reduce the precision of fit of the model. For the François' langurs dataset, the precision was 67.24%, primarily due to the disproportionately large number of negative samples, leading to 19 false positives, which lowered the precision. Nevertheless, the model performance yielded an F1 score of 80.41%, balancing precision and recall. Thus, while DeepADN performs robustly in detecting target species' calls, further optimization is required for cross-species applications to reduce false positives and enhance precision, particularly in diverse and acoustically complex and unbalanced dataset.

4.4 | Challenges of DeepADN

DeepADN significantly improved recall by detecting more true positive instances, especially under challenging acoustic conditions. However, its heightened sensitivity to the calls of white-headed langurs also increased false positives. This trade-off between recall and precision resulted in a slight decrease in overall F1 score performance, reflecting the challenge of balancing accurate detection of positive instances with minimizing misclassification of nontarget sounds.

The habitat of the white-headed langur is highly fragmented, characterized by heterogeneous and discontinuous patches of forest interspersed with large areas of farmland (Huang 2002; Huang et al. 2008). This fragmentation, combined with extensive human activities, generates significant noise pollution from agricultural practices, including machinery use, crop cultivation, and cargo transportation. Such noise pollution not only masks the calls of target species but also introduces nonbiological sounds that may lead to misclassification. These challenges are compounded by the complexity of the soundscape, as fragmented habitats often undermine the effectiveness of acoustic monitoring systems by creating masking effects that obscure the faint calls

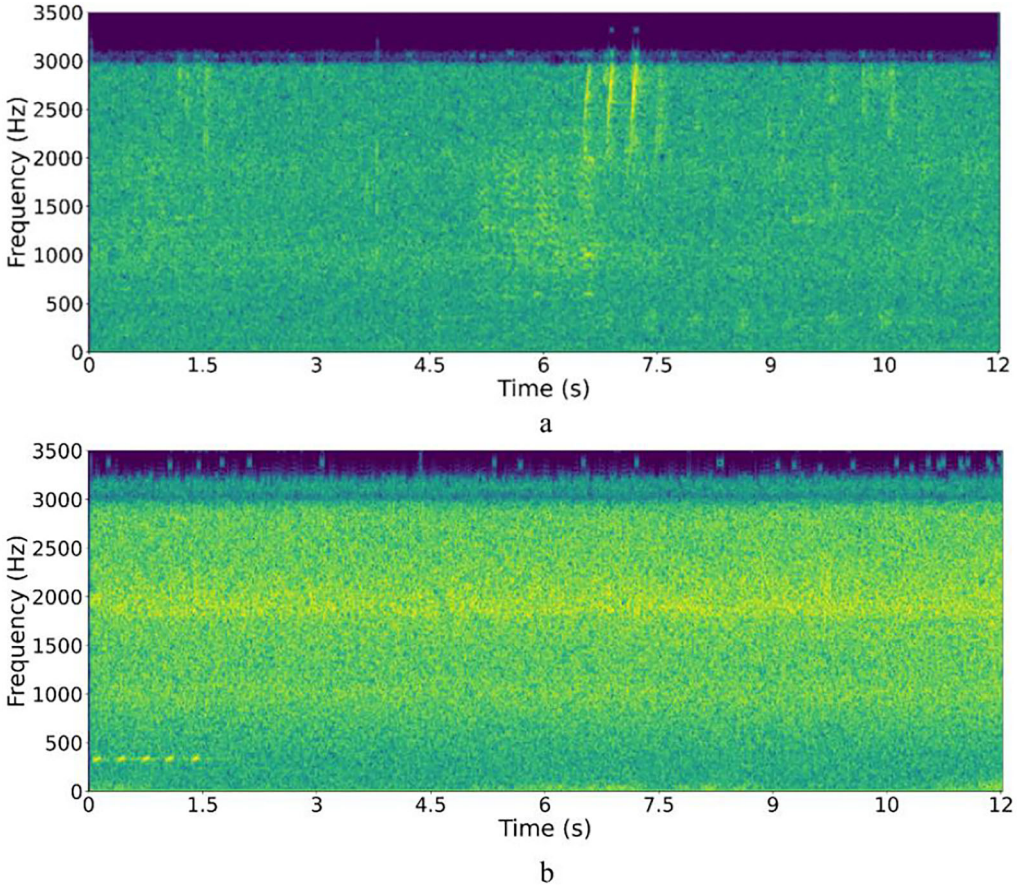

FIGURE 6 | Examples of false negatives and false positives: (a) a false positive sample and (b) a greater coucal call sample.

of white-headed langurs. False negatives were primarily caused by high-intensity background noise, which often masked the faint calls of white-headed langurs, especially in recordings taken from a distance (Figure 6a). The target calls were often inaudible amid environmental noise, making reliable detection difficult. In contrast, false positives frequently arose in acoustically complex environments where background noise included loud ambient sounds and calls from other species, such as the greater coucal [*Centropus sinensis* (Stephens, 1815)] (Figure 6b). These calls closely resembled adult male white-headed langur calls in frequency and temporal structure, with overlapping frequency ranges, harmonic patterns, and similar temporal rhythms. This posed significant challenges for the model, often resulting in misclassification.

The model's performance exhibits a pronounced dependence on environmental noise conditions. The model performs exceptionally well in clear recording environments, such as captive settings where François' langur calls are isolated from background noise. In these conditions, the distinct acoustic features of the calls allow the model to reliably identify target calls. In contrast, performance declines in noisy field environments (e.g., PAM data with wind, rain, or overlapping animal sounds), where background noise masks critical acoustic features of target calls, while vocalizations from other species closely resemble the target calls, leading to misclassification. This underscores the inherent trade-off between controlled-environment precision and ecological realism, emphasizing the need for adaptive noise-robust training strategies to bridge the gap between captivity and field deployment scenarios.

Adjusting the model's focus based on the duration of calls, inspired by the focal loss principle (Lin et al. 2017), can address these challenges. Adjusting the focal duration is particularly effective for addressing class imbalances, by down-weighting easy-to-classify examples and focusing on the hard-to-detect instances. In the case of false negatives, a duration-aware focal loss function improves the detection of longer and weaker calls that are often buried in place more weight on calls that have a longer duration and are more likely to belong to the target species, while down-weighting shorter, potentially irrelevant sounds.

5 | Conclusions

This paper presents an effective deep learning model for detecting adult male white-headed langur calls, using a combination of Fbank features and a CNN enhanced by the CBAM. Through PAM and advanced data processing techniques, DeepADN demonstrated high accuracy and robustness in noisy field conditions. This work addresses critical challenges in primate conservation by providing a noninvasive, scalable monitoring tool capable of tracking endangered primates in fragmented habitats. The model's adaptability was further validated through transfer learning, successfully identifying calls of François' langurs with limited training data. This adaptability illustrates the potential for

the model to be applied across various primate species, supporting broader conservation efforts. Additionally, our results highlight the value of combining resampling and data augmentation techniques, which enhance the model's ability to accurately identify primate calls despite dataset imbalances.

Future research should improve calls-specific duration robust loss functions to improve the detection of short-duration calls, which remain challenging for conventional neural networks. Overall, this study lays a foundation for the analysis of PAM technologies of endangered species, expanding the utility of the field of conservation bioacoustics.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

All DeepADN code and associated data are deposited at https://github.com/yhzdsk/DeepADN

Ethics Statement

All procedures followed relevant laws and regulations of the People's Republic of China and guidelines approved by the Guangxi Normal University's Ethical Review and Academic Committee (No.202412-005).

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supplementary Materials: inz270019-sup-0001-SuppMat.docx